

Elizabeth L. Stippell

(516)-974-9565

stippell@sas.upenn.edu

<https://liz-stippell.github.io/>

EDUCATION

University of Southern California

PhD: Physical/Theoretical Chemistry

- Oleg Prezhdo group

Los Angeles, CA

08/2021 – 12/2025

University at Buffalo

Bachelor of Science, Magna Cum Laude, Chemistry

- Minor in Mathematics
- Alexey Akimov research group

Buffalo, NY

08/2017 – 05/2021

RESEARCH EXPERIENCE

UNIVERSITY OF PENNSYLVANIA

01/2026 – Present

Professor Abraham Nitzan Research Group

- Developing a model of molecular synchronization induced by photons in the realm of vibrational strong coupling, emphasizing the role of collective effects in optical cavities for chemical reactivity and related phenomena
- Member of The Society of Fellows in Public Knowledge

2026 – 2027

UNIVERSITY OF SOUTHERN CALIFORNIA

Professor Oleg Prezhdo Research Group

08/2021 – 12/2025

- Research focus on a fundamental understanding of energy materials for solar cell design including perovskites and quantum dots using molecular dynamics/non-adiabatic molecular dynamics simulations
- Mentor to graduate students

LOS ALAMOS NATIONAL LABORATORY

T-1 & T-4 Divisions

06/2022 – 12/2025

- Developing machine learning methods to accurately predict and prevent chemical warfare agents
- Constructed a machine learning active learning potential to deepen understanding of nuclear fuels via molecular dynamics simulations

PUBLICATIONS (Newest to Oldest)

First Author Publications

- The Quantum Glissando Effect: Expanding the particle in a box model to include nonadiabatic effects
- **Stippell, E.;** Mora Perez, C.; Favate, N.; Huang, L.; Li, C. W.; Prezhdo, O. V.
Computational Screening of Ligands for Enhanced Interactions between Lead Halide Perovskite Quantum Dots. *J. Phys. Chem. Lett.* **2025**, 16 (23), 5666–5673. <https://doi.org/10.1021/acs.jpcllett.5c01307>.

In Preparation

-
- **Stippell, E.**; Li, W.; Quarti, C.; Beljonne, D.; Prezhdo, O. V.
Enhancing Interlayer Charge Transport of Two-Dimensional Perovskites by Structural Stabilization via Fluorine Substitution. *ACS Appl. Mater. Interfaces* **2025**, 17 (1), 2032–2040.
<https://doi.org/10.1021/acsami.4c17876>.
 - **Stippell, E.**; Alzate-Vargas, L.; Subedi, K. N.; Tutchton, R. M.; Cooper, M. W. D.; Tretiak, S.; Gibson, T.; Messerly, R. A.
Building a DFT+U Machine Learning Interatomic Potential for Uranium Dioxide. *Artificial Intelligence Chemistry* **2024**, 2 (1), 100042.
<https://doi.org/10.1016/j.aichem.2023.100042>.
 - **Stippell, E.**; Akimov, A. V.; Prezhdo, O. V.
PySyComp: A Symbolic Python Library for the Undergraduate Quantum Chemistry Course. *J. Chem. Educ.* **2023**, 100 (10), 4077–4084.
<https://doi.org/10.1021/acs.jchemed.2c00974>.

Co-Authored Publications

- Xu, X.; Han, X.; **Stippell, E.**; Fang, W.-H.; Prezhdo, O. V.; Long, R.
Anisotropic Exciton-Polariton Relaxation and Phonon Bottleneck in Microcavity-Confined Metal Halide Perovskites: Ab Initio Quantum Dynamics. *Nano Lett.* **2025**, 25 (34), 13063–13069.
<https://doi.org/10.1021/acs.nanolett.5c03697>.
- Zhang, P.; **Stippell, E.**; Hou, Z.; Prezhdo, O. V.; Li, W.
Mitigating Band Tailing in Kesterite Solar Absorbers: Ab Initio Quantum Dynamics. *J. Am. Chem. Soc.* **2024**, 146 (46), 32147–32157.
<https://doi.org/10.1021/jacs.4c14416>.
- Ma, X.; Tian, X.; **Stippell, E.**; Prezhdo, O. V.; Long, R.; Fang, W.-H.
Self-Passivation of Halide Interstitial Defects by Organic Cations in Hybrid Lead-Halide Perovskites: Ab Initio Quantum Dynamics. *J. Am. Chem. Soc.* **2024**, 146 (42), 29255–29265.
<https://doi.org/10.1021/jacs.4c12634>.
- Shakiba, M.; **Stippell, E.**; Li, W.; Akimov, A. V.
Nonadiabatic Molecular Dynamics with Extended Density Functional Tight-Binding: Application to Nanocrystals and Periodic Solids. *J. Chem. Theory Comput.* **2022**, 18 (9), 5157–5180.
<https://doi.org/10.1021/acs.jctc.2c00297>.

RESEARCH AWARDS

- | | |
|--|-------------|
| • Belgian American Educational Foundation Fellowship Recipient | 2023-2024 |
| • Milligan Fellowship Recipient | Summer 2021 |

UNIVERSITY OF SOUTHERN CALIFORNIA

-
- | | |
|--|-------------|
| • Graduate School Fellowship Recipient | 2022 - 2023 |
|--|-------------|

UNIVERSITY AT BUFFALO

- | | |
|---|-------------|
| • Western New York American Chemical Society Award | 2021 |
| • American Chemical Society Division of Inorganic Chemistry Award | 2021 |
| • Provost Scholarship Recipient | 2017 - 2021 |
| • Albert Padwa Summer Research Award | Summer 2020 |

PRESENTATIONS

- | | |
|---|---------|
| • Guest Lecture: Ithaca College Chemistry Department | 09/2026 |
| • The American Chemical Society (ACS) Fall 2025 Conference: Energy and Fuels Division | 08/2025 |
| • The American Chemical Society (ACS) Fall 2025 Conference: Physical Chemistry Division | 08/2025 |
| • USC Women in Science and Engineering STEMBytes Seminar | 03/2025 |
| • Virtual International Seminar on Theoretical Advancements (VISTA) | 02/2025 |
| • Virtual Theoretical Division Lightning Talk Series (Los Alamos National Laboratory) | 07/2023 |

WORKSHOPS ATTENDED

- | | |
|---|-------------|
| • Telluride School on Theoretical Chemistry | Summer 2025 |
| • Compchem Cybertraining Workshop | Summer 2023 |
| • Libra Winter Workshop | Winter 2022 |
| • Compchem Cybertraining Workshop | Summer 2021 |

TEACHING (TA) EXPERIENCE

- | | |
|---|-------------|
| • CHEM102 (The Molecular World – General Chemistry) | Fall 2023 |
| • CHEM115b (Advanced General Chemistry) | Spring 2022 |
| • CHEM322a (Organic Chemistry 1) | Fall 2021 |